

1. Approach and Performance Summary

Model	Precision	Recall	F1-Score	AUC	Time (s)
LogisticRegression	0.96	0.96	0.96	0.9757	126.13
GloVe + LR	0.93	0.89	0.90	0.9478	186.76
Transformers (toxic-bert)	0.98	0.98	0.98	0.9955	2776.53
Zero-shot	0.92	0.45	0.54	0.9316	59.71
Role prompting	0.92	0.12	0.07	0.8339	68.01
Few-shot	0.88	0.73	0.78	0.8217	199.62

Note: The best model from Project 1 was Logistic Regression trained on TF-IDF/Keyword features.

2. Comparative Analysis

2.1 Predictive Performance

Best model: Transformers (toxic-bert) - **F1-Score:** 0.98 (excellent precision/recall balance) - **AUC:** 0.9955 (near-perfect discrimination) - Model specialized for toxicity detection

Word Embeddings (GloVe + LR): - Intermediate performance ($F1 = 0.90$) - High precision (0.93) and high recall (0.89) - Many false negatives

Prompt Engineering: - Disappointing performance ($F1 < 0.80$) - Very low recall in role prompting (0.12) - Few-shot improves results ($F1 = 0.78$) but not significantly

2.2 Computation Time

Fastest: Zero-shot (59.71s) and Role prompting (68.01s)

Intermediate: GloVe + LR (186.76s) and Few-shot (199.62s)

Slowest: Transformers (2777s, approximately 46 minutes)

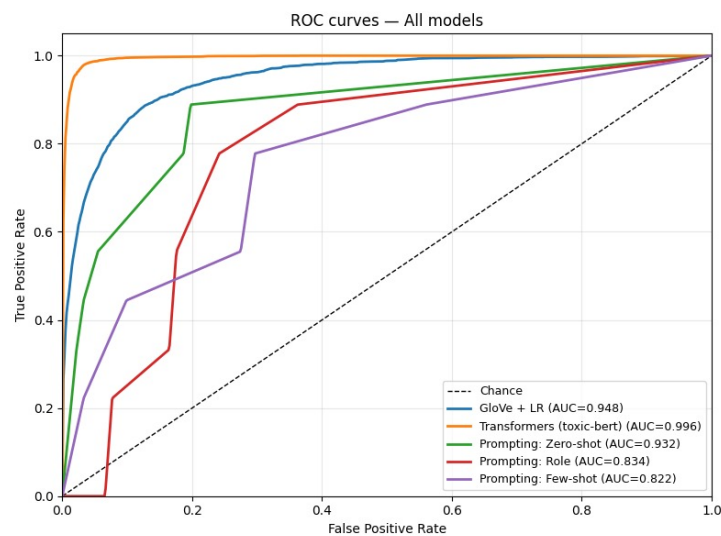


Figure 1: ROC curves for all models. The Transformers (toxic-bert) model achieves the best performance with AUC close to 1.

3. Critical Reflections

3.1 Computational Cost

Impact on deployment: - **Transformers:** Excellent performance but prohibitive cost for processing large volumes in real-time. Solution: use distilled models or batch processing. - **Embeddings:** Reasonable compromise between speed and performance. - **Prompting:** Fast but insufficient performance and potentially high API costs.

3.2 Bias

Example: "She is a woman" flagged as toxic

Possible causes: - **Training bias:** If the corpus contained sexist comments associating "woman" with negative contexts - **Biased co-occurrences:** The model learns statistical correlations without understanding context - **Over-generalization:** Certain demographic keywords are over-represented in toxic examples

Risks: Unfair moderation, censorship of legitimate gender discussions, amplification of stereotypes.

3.3 Transparency

More explainable: GloVe + Logistic Regression - Can inspect feature weights - Identification of influential words possible

Less explainable: Transformers - Millions of parameters, complex attention mechanisms - Requires interpretation tools (SHAP, attention visualization)

Intermediate: Prompting - Prompt is visible but the LLM's internal reasoning remains opaque

3.4 Confidentiality

Risks with external LLM APIs: - **Data exposure:** Comments are sent to third-party servers (OpenAI, Anthropic...) - **Logs and retention:** Risk of prolonged storage of user content - **GDPR non-compliance:** Transfer of personal data outside the EU - **Consequences for Wikipedia:** Violation of contributor privacy, loss of community trust

Solutions: On-premise models, prior anonymization, strict contractual clauses.

4. Recommendations

1. **Production:** Use **toxic-bert** with optimized infrastructure (GPU, batching) 2. **Rapid prototyping:** **GloVe + LR** for an MVP 3. **Avoid:** Prompt engineering for this task (insufficient performance, API costs/risks) 4. **Monitoring:** Regular bias audits, testing on demographic subgroups 5. **Transparency:** Develop an explainability dashboard for moderators

Conclusion

Transformers offer the best performance but require significant resources. The final choice depends on the project's latency constraints, budget, and confidentiality requirements.